

IB Mathematics — Volume of Revolution

10 Original Paper 2 Style Questions | AAHL & AIHL

Instructions: Show all relevant working. Give exact answers where appropriate and numerical answers to the stated accuracy. A calculator may be used where appropriate. This is an original practice set and is not an official IB examination paper.

Question 1 [7 marks]

The region bounded by the curve $y = x^2$, the x-axis, $x = 0$ and $x = 2$ is rotated through 360° about the x-axis.

(a) Write down an integral representing the volume of the solid formed. [2]

(b) Find the exact volume of the solid. [5]

Question 2 [8 marks]

The region bounded by $y = 4 - x^2$ and the x-axis is rotated through 360° about the x-axis.

(a) Find the x-coordinates of the points where the curve meets the x-axis. [2]

(b) Hence, find the exact volume of revolution. [6]

Question 3 [9 marks]

The region bounded by $y = x + 2$, $y = 0$, $x = -1$ and $x = 2$ is rotated through 360° about the x-axis.

(a) Sketch the region and indicate the axis of rotation. [2]

(b) Find the exact volume of the solid formed. [7]

Question 4 [9 marks]

The region enclosed by $y = \sqrt{x}$, $y = x/2$ and the x-axis is rotated through 360° about the x-axis.

- (a) Find the coordinates of the non-origin point of intersection of the two curves. [2]
- (b) Explain why washers are not required for the rotation about the x-axis. [1]
- (c) Find the exact volume of the solid formed. [6]

Question 5 [10 marks]

The region bounded by $y = x^2$ and $y = 2x$ is rotated through 360° about the x-axis.

- (a) Find the points of intersection of the curves. [3]
- (b) Determine which curve forms the outer radius for $0 < x < 2$. [1]
- (c) Find the exact volume of the resulting solid using an integral. [6]

Question 6 [10 marks]

The region between the curves $y = x + 1$ and $y = x^2$ is rotated through 360° about the x-axis.

- (a) Find the points of intersection of the curves. [2]
- (b) Show that the volume can be represented by an integral involving the difference of two squared functions. [2]
- (c) Find the exact volume of the solid. [6]

Question 7 [10 marks]

The region enclosed by $y = 4 - x^2$ and $y = 2$ is rotated through 360° about the line $y = 2$.

- (a) Find the x-coordinates of the intersection points. [2]
- (b) Write down the radius of a typical disc in terms of x . [2]
- (c) Hence, find the exact volume of the solid formed. [6]

Question 8 [11 marks]

The region bounded by $y = x^2$, $y = 4$ and $x = 0$ is rotated through 360° about the y -axis.

- (a) Express x in terms of y for the boundary $x = \sqrt{y}$. [1]
- (b) Write down an integral with respect to y for the volume. [3]
- (c) Find the exact volume of the solid. [7]

Question 9 [11 marks]

The region bounded by $y = x + 1$, $y = 1$ and $x = 3$ is rotated through 360° about the y -axis.

- (a) Express x in terms of y for the sloping boundary. [2]
- (b) State the limits of integration if the volume is found with respect to y . [2]
- (c) Find the exact volume of the solid. [7]

Question 10 [12 marks]

The curve $y = e^{-x}$ and the x-axis enclose a region for $0 \leq x \leq 3$. This region is rotated through 360° about the x-axis.

- (a) Write down an integral for the volume. [2]
- (b) Use integration to obtain an exact expression for the volume. [4]
- (c) Give the volume correct to 3 significant figures. [2]
- (d) The same region is instead rotated about the line $y = 1$. Explain why the volume cannot be found by simply replacing y with $1 - y$ in the original volume formula, and set up a correct integral for the new volume. [4]
